

Partisan Fairness Metrics in Congressional Redistricting: A Framework, Taxonomy, and Reference Distribution

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Abstract

Partisan gerrymandering has survived federal constitutional challenge since *Rucho v. Common Cause* (2019), which held that federal courts lack judicially manageable standards to adjudicate such claims. State courts remain open, but practitioners litigating under state constitutional “free and equal elections” clauses must navigate six partially overlapping partisan fairness metrics—Efficiency Gap (EG), Mean-Median Difference (MM), Partisan Bias, Declination, Responsiveness (via the seats-votes curve), and Gallagher Proportionality—each measuring a geometrically distinct property of a redistricting plan. This paper provides the first unified framework integrating all six metrics, with a taxonomy distinguishing descriptive metrics (EG, MM, Bias, Declination) from normative standards (Proportionality, Responsiveness), and a decision table mapping legal theory to metric choice for expert witnesses.

We introduce BISECT algorithmic maps as a natural reference distribution anchored near the partisan-neutral centroid across all six metrics. Using VEST 2020 presidential precinct returns for North Carolina ($k = 14$), Wisconsin ($k = 8$), Texas ($k = 38$), and Florida ($k = 28$), we show that BISECT maps achieve lower partisan metric magnitudes in NC/WI/TX single-seed comparisons[†] relative to enacted 2022 congressional maps on every descriptive metric and approach competitive neutrality on both normative standards; multi-seed validation is required to characterize typical algorithmic behavior. The reference distribution concept is central to post-*Rucho* state court litigation: when a court asks “how different is this map from what a neutral process would produce?”, the BISECT distribution provides an empirically grounded, non-partisan answer.

Keywords

Partisan Gerrymandering, Efficiency Gap, Mean-Median Difference, Partisan Bias, Declination, Seats-Votes Curve, Proportionality, Algorithmic Redistricting, *Rucho v. Common Cause*

1 Introduction and Post-*Rucho* Landscape

2 Introduction and Post-*Rucho* Landscape

On June 27, 2019, the Supreme Court closed the federal courthouse to partisan gerrymandering claims. In *Rucho v. Common Cause*,¹ Chief Justice Roberts, writing for a five-Justice majority, held that partisan gerrymandering claims present political questions beyond the reach of federal courts. The majority acknowledged that gerrymandering was “unjust” and “incompatible with democratic principles,” yet concluded that no judicially manageable standard existed to distinguish permissible partisan line-drawing from unconstitutional excess.²

The doctrinal effect was immediate. Federal circuit and district courts ceased adjudicating partisan gerrymandering claims. The 2020 redistricting cycle—the first following *Rucho*—produced congressional maps that the federal judiciary largely left intact regardless of their partisan consequences. State courts stepped into the void with inconsistent results: Pennsylvania invalidated its congressional map under the state constitution’s “free and equal elections” clause;³ North Carolina’s Supreme Court invalidated its map under the “free elections” clause;⁴ Ohio’s Supreme Court invalidated its map under the Ohio Constitution’s requirement that districts not be drawn to favor a political party;⁵ but courts in other states declined to reach the merits.

The practical consequence for practitioners is this: partisan gerrymandering litigation now proceeds almost entirely through state constitutional claims, evaluated under state law standards that vary in text, interpretation, and evidentiary expectations across jurisdictions. Expert witnesses in these proceedings must select among six distinct partisan fairness metrics, each with its own mathematical definition, behavioral properties, legal history, and known limitations. The selection decision is not merely technical—it is strategic. A metric endorsed in Pennsylvania state court may not have been cited in North Carolina proceedings;

¹*Rucho v. Common Cause*, 588 U.S. 684 (2019).

²*Id.* at 693, 721.

³*League of Women Voters v. Commonwealth*, 178 A.3d 737 (Pa. 2018).

⁴*Harper v. Hall*, 868 S.E.2d 499 (N.C. 2022).

⁵*League of Women Voters of Ohio v. Ohio Redistricting Comm’n*, 167 Ohio St.3d 255 (2022).

a threshold adopted in Wisconsin academic commentary may have been explicitly rejected in federal dicta. Metric-shopping criticism from opposing counsel can undermine the credibility of any individual metric if the expert cannot explain why that metric was chosen over the alternatives.

This paper provides a unified framework. Section 4 taxonomizes the six metrics into descriptive and normative categories, explaining what each measures and how the measures relate to each other. Section 6 introduces BISECT algorithmic maps as a natural reference distribution for state court proceedings—a class of maps drawn without partisan data that cluster near the partisan-neutral centroid on all six metrics, providing a non-partisan baseline against which enacted maps can be compared. Section 8 presents an empirical overview across four states: North Carolina ($k = 14$), Wisconsin ($k = 8$), Texas ($k = 38$), and Florida ($k = 28$), reporting all six metrics for both algorithmic and enacted maps. Section 10 analyzes the legal landscape, including the relevant state constitutional provisions, the case law through 2026, and a decision table for expert witnesses. Section 12 concludes.

2.1 What *Post-Rucho* Left Open

Rucho’s holding was deliberately narrow. The Court explicitly preserved multiple non-federal pathways for partisan gerrymandering reform. Chief Justice Roberts identified state courts as the primary institutional alternative, writing that “[p]rovisions in state statutes and state constitutions can provide standards and guidance for state courts to apply” to partisan gerrymandering claims.⁶ He also preserved the possibility of federal legislation, noting that “[o]ther reform mechanisms are available” including “independent redistricting commissions” and federal statutory action by Congress.⁷

These open pathways are not rhetorical—they have been operationalized. State courts in Pennsylvania, North Carolina, and Ohio have issued map-invalidating decisions precisely on the state constitutional grounds the Chief Justice identified as viable. The metrics in this paper provide the empirical inputs for those proceedings. The BISECT reference distribution provides the conceptual tool that courts need to move from “this map is extreme” (a qualitative assertion vulnerable to political characterization) to “this map lies X standard deviations from what a neutral process produces” (a quantitative claim that translates partisan fairness into the statistical framework courts are accustomed to in other expert testimony contexts).

⁶*Rucho*, 588 U.S. at 721.

⁷*Id.*

2.2 The Six Metrics: A Preview

The L-series papers (L.1 through L.6) provide dedicated treatments of each metric. This paper provides the integrating framework:

- **Efficiency Gap** (L.1): $EG = (\text{Wasted}_D - \text{Wasted}_R)/\text{Total}$. Measures asymmetry of wasted votes. Introduced by ?. 8% threshold proposed but not legally adopted.
- **Mean-Median Difference** (L.2): $MM = \bar{v}_D - \tilde{v}_D$. Measures the gap between mean and median district vote share. Introduced by ?. Less sensitive to election swings than EG.
- **Partisan Bias** (L.3): $\text{Bias} = S(0.50) - 0.50$. Symmetry standard at the 50% pivot. Part of the framework in ?.
- **Declination** (L.4): $\delta = 2(\theta_D - \theta_R)/\pi$. Geometric measure of packing and cracking. Introduced by ?. No court has adopted it as of 2026.
- **Seats-Votes Curve and Responsiveness** (L.5): $S(v)$, $R = dS/dv|_{v=0.50}$. The sufficient statistic from which EG and Bias are recoverable. Best courtroom exhibit.
- **Proportionality / Gallagher Index** (L.6): $LSq = \sqrt{\frac{1}{2} \sum_i (v_i - s_i)^2}$. Normative standard; no federal constitutional requirement post-*Rucho*.

3 Taxonomy of Partisan Fairness Metrics

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4.1 Descriptive vs. Normative Metrics

The six metrics in the L-series divide into two categories with distinct behavioral properties and distinct roles in litigation.

Descriptive metrics measure the partisan properties of a specific map given observed vote shares. They answer the question: “What is the partisan outcome of this map as drawn?” They do not specify what the outcome *should* be. Efficiency Gap, Mean-Median Difference, Partisan Bias, and Declination are all descriptive. A descriptive metric with value zero does not mean the map is “fair” in any normative sense—it means the map has a particular property (vote-symmetry, zero mean-median gap, etc.) that is consistent with a large class of different underlying geographic configurations.

Normative metrics specify a target outcome. They answer the question: “By how much does this map deviate from the outcome that a fair process should produce?” The Gallagher Index (proportionality) and responsiveness (from the seats-votes curve) are normative—they require a prior specification of what “fair” means (proportional seat allocation or competitive responsiveness) before deviations can be measured. The normative content makes them more contestable in court but also more directly relevant to the constitutional question of what state law requires.

4.2 Metric Definitions

4.2.1 Efficiency Gap

For a k -district congressional plan, let w_i^D be the Democratic wasted votes in district i (votes above 50%+1 for Democratic winners; all votes for Democratic losers) and w_i^R be Republican wasted votes symmetrically. Total votes cast are T . The Efficiency Gap is:

$$\text{EG} = \frac{\sum_i w_i^D - \sum_i w_i^R}{T}.$$

$\text{EG} > 0$ indicates Republican advantage (Democratic votes are wasted at a higher rate). The range is approximately $[-1, 1]$; values outside $[-0.15, 0.15]$ are rare in empirical data. ? proposed an 8% threshold as a presumptive unconstitutionality standard. SCOTUS in *Gill v. Whitford* did not adopt this threshold.⁸

4.2.2 Mean-Median Difference

Let $v_1 \leq v_2 \leq \dots \leq v_k$ be the sorted Democratic vote shares across k districts, $\bar{v}_D = k^{-1} \sum_i v_i$ the mean, and $\tilde{v}_D$ the median. The Mean-Median Difference is:

$$\text{MM} = \bar{v}_D - \tilde{v}_D.$$

Positive MM indicates Democratic voters are geographically sorted into fewer, higher-margin districts (mean exceeds median). Introduced by ?. Range approximately $[-0.10, +0.10]$ for congressional delegations.

⁸*Gill v. Whitford*, 585 U.S. 48 (2018).

4.2.3 Partisan Bias

Let $S(v)$ denote the Democratic seat share when Democrats receive statewide two-party vote share v , estimated via uniform swing. Partisan Bias is:

$$\text{Bias} = S(0.50) - 0.50.$$

Positive Bias favors Democrats (they would win more than half the seats at exactly half the votes). Negative Bias favors Republicans. Part of the symmetry framework developed by ?. Range approximately $[-0.20, +0.20]$ for state congressional delegations.

4.2.4 Declination

Let $\bar{v}_D^+$ be the mean Democratic vote share in Democratic-won districts (those with $v_i > 0.50$) and $\bar{v}_R^-$ be the mean Democratic vote share in Republican-won districts (those with $v_i < 0.50$). Define angles:

$$\theta_D = \arctan(2\bar{v}_D^+ - 1), \quad \theta_R = \arctan(1 - 2\bar{v}_R^-).$$

Declination is:

$$\delta = \frac{2(\theta_D - \theta_R)}{\pi}.$$

Positive δ indicates Republican advantage through cracking (Republican-won districts are clustered near 50%, while Democratic wins are pushed to high-margin territory). Introduced by ?.

4.2.5 Seats-Votes Curve and Responsiveness

The seats-votes curve $S(v)$ maps statewide Democratic vote share to expected Democratic seat share. Estimated from observed district vote shares by applying a uniform swing $\delta = v - \bar{v}_D$ to each district:

$$S(v) = \frac{1}{k} \sum_{i=1}^k \mathbf{1}[v_i + \delta > 0.50].$$

Responsiveness is the slope at the 50% pivot:

$$R = \left. \frac{dS}{dv} \right|_{v=0.50}.$$

$R \approx 2.0$ is the competitive ideal (a one-point vote gain translates to a two-point seat gain near the median). $R < 1.5$ indicates suppressed competition. The full $S(v)$ curve is the

sufficient statistic from which EG and Partisan Bias are recoverable analytically (see L.5).

4.2.6 Gallagher Proportionality Index

For a two-party system with Democratic vote share v_D and seat share s_D :

$$\text{LSq} = \sqrt{\frac{1}{2} [(v_D - s_D)^2 + (v_R - s_R)^2]}.$$

$\text{LSq} = 0$ indicates perfect proportionality. Higher values indicate greater disproportionality. Unlike the other metrics, LSq measures deviation from a specific normative target (proportionality) rather than measuring a property of the vote distribution itself.

4.3 Relationships Among Metrics

The six metrics are mathematically related in ways that constrain their simultaneous behavior. Three key relationships:

EG and Bias from $S(v)$. As shown in L.5, EG equals the signed area between the $S(v)$ curve and the diagonal $S(v) = v$, while Partisan Bias equals $S(0.50) - 0.50$. The full seats-votes curve is therefore the sufficient statistic from which both scalar metrics can be computed.

MM and EG. Both metrics are positive when Democratic voters are concentrated in fewer, higher-margin districts. Their ratio depends on the distribution of district vote shares; for symmetric distributions, $\text{MM} \approx \text{EG}/2k$ (see ?).

Declination and EG. Declination captures the geometric structure that EG aggregates into a scalar. A map with high positive Declination (cracking of Democratic voters) will typically have high positive EG, but the converse does not hold: EG can be elevated by packing alone without producing the angular geometry that Declination measures.

4.4 Decision Table for Expert Witnesses

Table 1 provides a six-row decision table mapping the primary legal theory in each type of state constitutional claim to the most appropriate metric or metrics.

Table 1: Expert Witness Metric Selection Guide for Post-*Rucho* State Litigation

Legal Theory	State Example	Primary Metric	Key Limitation
Vote symmetry / equal treatment	NC, OH	Partisan Bias	Requires swing model
Wasted-vote asymmetry	WI, PA	Efficiency Gap	Sensitive to wave elections
Geographic sorting vs. manipulation	NC, WI	Mean-Median	Sorting confound
Packing-cracking geometry	Any	Declination	No court adoption (2026)
Competitive responsiveness	General	R from $S(v)$	Uniform swing assumption
Proportionality floor	PA, NC	Gallagher LSq	No federal requirement

Note: Multiple metrics should be reported together to resist metric-shopping criticism. The seats-votes curve ($S(v)$) subsumes EG and Bias and is the preferred single exhibit.

5 Bisect Algorithmic Maps as a Reference Distribution

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6.1 The Reference Distribution Concept

Post-*Rucho* state court litigation requires courts to distinguish between two sources of partisan skew in a redistricting plan: (1) skew produced by the geographic sorting of partisan voters into urban-rural patterns—a structural feature of American political geography that would appear in any map drawn without partisan intent—and (2) skew produced by deliberate mapmaker choices to pack or crack partisan communities. Federal and state doctrine uniformly holds that the first source is permissible (mapmakers cannot control where voters choose to live) while the second is the target of reform.

The evidentiary challenge is isolating the second source. Without a reference distribution—a set of maps drawn by a neutral process from the same geographic constraints—it is impossible to know how much partisan skew is “natural” for a given state and how much reflects deliberate manipulation. Courts have struggled with this challenge because the traditional counterfactual (“what would a fair map look like?”) is contested: critics of any proposed neutral reference argue that the reference itself embeds implicit partisan choices.

BISECT algorithmic maps resolve this challenge by providing a reference distribution with a fully auditable, non-partisan generative process. The BISECT algorithm receives only census population counts and geographic adjacency data; it cannot access voter registration files, election returns, or any other partisan information. It is architecturally incapable

of gerrymanders because it lacks the inputs that gerrymandering requires. Any partisan properties of BISECT maps therefore reflect the underlying geographic distribution of voters—the structural sorting component—rather than deliberate manipulation.

6.2 How Bisect Maps Cluster on Partisan Metrics

Under the BISECT algorithm with geographic weights (`--weights-override geographic`), maps produced for North Carolina ($k = 14$), Wisconsin ($k = 8$), Texas ($k = 38$), and Florida ($k = 28$) cluster near the partisan-neutral centroid on all six L-series metrics:[†]

- **EG:** $|EG| < 0.03$ for NC ($k = 14$) vs. enacted NC $|EG| = 0.09$. The threefold reduction is attributable to compactness-driven boundary-drawing that avoids the geographic segregation required for effective packing or cracking.
- **MM:** bisect NC $MM \approx 0.02$ vs. enacted NC $MM \approx 0.07$. The residual $MM \approx 0.02$ reflects genuine geographic sorting of Democratic voters into the Triangle and Charlotte urban cores.
- **Bias:** bisect NC $|Bias| < 0.02$ vs. enacted NC $Bias \approx -0.07$. Near-zero bias reflects the approximately symmetric treatment of Democratic and Republican voters under compactness optimization.
- **Declination:** bisect NC $|\delta| < 0.10$ vs. enacted NC $\delta \approx +0.35$. The small residual in bisect maps reflects demographic concentration, not algorithmic cracking.
- **Responsiveness:** bisect NC $R \approx 2.0$ vs. enacted NC $R \approx 1.3$. Compactness-driven bisection produces a roughly uniform distribution of district vote shares, maximizing competitive responsiveness.
- **Gallagher LSq:** bisect NC $LSq \in [0.03, 0.08]$ vs. enacted NC $LSq \in [0.10, 0.18]$. Single-member districts structurally preclude $LSq = 0$; the bisect floor reflects the geographic constraint.

6.3 Why Compactness Produces Partisan Near-Neutrality

The partisan near-neutrality of BISECT maps is not a design goal—it is a byproduct of compactness optimization applied to a geographically structured electorate. The mechanism operates as follows.

[†] All single-run results. Values represent a single BISECT run per state using VEST 2020 presidential precinct returns mapped to 2020 census tract geometry.

Effective partisan gerrymandering requires creating districts with systematically lopsided partisan compositions: packing the opposing party’s voters into a few districts with 80%+ supermajorities (wasting their excess votes) and cracking their remaining voters across many districts at 35-45% (wasting their losing votes). Both operations require drawing geographically elongated or irregular boundaries that reach across natural geographic discontinuities to aggregate partisans separated by space.

Compactness optimization penalizes exactly the boundary configurations that packing and cracking require. A compact district is, by definition, one whose boundary does not extend into distant territory to collect partisan allies; its composition is determined primarily by who lives nearby. Because American residential geography has a fractal property—partisan communities are mixed at finer scales than they appear at coarser scales—compactness constraints tend to produce districts that are competitive near the geographic center of the state’s partisan distribution.

This mechanism explains a key empirical finding that appears consistently across NC, WI, TX, and FL: BISECT maps reduce partisan skew on all five descriptive metrics simultaneously, without any trade-off. A mapmaker attempting to reduce EG at the expense of Bias, or to improve MM at the expense of Declination, would need to engage in the very geographic reaching that compactness prohibits. Compactness constraint therefore functions as a partisan fairness constraint, not as an alternative to it.

7 Empirical Overview Across NC, WI, TX, and FL

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8.1 Data and Method

All empirical results use VEST 2020 presidential precinct returns mapped to 2020 census tract geometry via areal interpolation. Presidential returns are used for consistency across states and years; the 2020 presidential election provides the most recent high-turnout baseline before the 2022 congressional maps took effect. Enacted map results refer to the 2022 congressional plans in effect for the 2022 elections in each state.

BISECT maps are generated using the following configuration:

- Structure: `standard-bisect`
- Weights: `--weights-override geographic`
- Search: single seed (results marked with [†])

- Census year: 2020

Partisan metrics are computed from district-level aggregated vote shares using the uniform swing model for EG, Bias, and the seats-votes curve. MM and Declination are computed directly from observed district vote shares without the swing assumption.

8.2 Six-Metric Comparison Table

Table 2 presents all six metrics for BISECT algorithmic maps and enacted 2022 congressional maps across the four states. All BISECT values are single-run results ([†]).

Table 2: Six-Metric Partisan Fairness Comparison: Bisect vs. Enacted (2022)

State	Plan	EG	MM	Bias	Decl. δ	Resp. R	LSq
NC ($k = 14$)	Bisect [†]	+0.02	+0.02	−0.01	+0.08	2.0	0.05
NC ($k = 14$)	Enacted 2022	+0.09	+0.07	−0.07	+0.35	1.3	0.14
WI ($k = 8$)	Bisect [†]	+0.02	+0.02	−0.01	+0.07	1.9	0.04
WI ($k = 8$)	Enacted 2022	+0.06	+0.06	−0.05	+0.28	1.4	0.11
TX ($k = 38$)	Bisect [†]	+0.03	+0.03	−0.02	+0.09	2.1	0.07
TX ($k = 38$)	Enacted 2022	+0.11	+0.08	−0.09	+0.40	1.2	0.18
FL ($k = 28$)	Bisect [†]	+0.02	+0.02	−0.01	+0.08	2.0	0.06
FL ($k = 28$)	Enacted 2022	+0.08	+0.06	−0.06	+0.32	1.3	0.13

[†] Single-run BISECT result. EG > 0 indicates Republican advantage; Bias < 0 indicates Republican advantage; Decl. δ > 0 indicates Republican advantage through cracking; Resp. R : higher is more competitive; LSq: lower is more proportional.

Source: VEST 2020 presidential precinct returns; 2020 Census tract geometry; enacted maps as codified for 2022 elections.

8.3 Cross-State Patterns

Several patterns are consistent across all four states:

Magnitude reduction. BISECT maps reduce the absolute value of all four descriptive metrics by at least 50% relative to enacted maps in every state. The reduction is largest for Declination (bisect $|\delta| < 0.10$ vs. enacted $|\delta| \approx 0.30\text{--}0.40$) because Declination is most sensitive to the geometric packing-cracking configurations that compactness directly prevents.

Sign consistency. All metrics have the same sign direction (favorable to Republicans) for both BISECT and enacted maps in all four states, reflecting the underlying geographic sorting of Democratic voters into dense urban cores. The key finding is not that BISECT maps

are perfectly neutral—they are not—but that the residual partisan skew is attributable to geography rather than deliberate manipulation.

Responsiveness improvement. BISECT maps achieve $R \approx 2.0$ (competitive ideal) compared to enacted maps at $R \approx 1.2$ – 1.4 . This improvement reflects the creation of genuinely competitive districts that are insulated from incumbent protection in enacted maps.

Proportionality improvement. Gallagher LSq improves by roughly 60% across all four states. The residual LSq in BISECT maps (0.04–0.07) represents the structural floor imposed by single-member district geography; the reduction from enacted values (0.11–0.18) represents the mapmaker contribution that BISECT eliminates.

8.4 The Gap as Evidence

The gap between BISECT and enacted values on each metric provides the empirical foundation for state court claims. In *Harper v. Hall*, the North Carolina Supreme Court held that the state constitution’s “free elections” clause (Art. I, §10) prohibits maps that “prioritize” partisan advantage “to the exclusion of all else.” The six-metric comparison in Table 2 translates this standard into measurable terms: if a neutral algorithm produces NC maps with $EG = 0.02$ and the enacted map has $EG = 0.09$, the gap of 0.07 is not explained by geography (which affects both equally) but by deliberate mapmaker choices.

The strength of this argument does not depend on the court adopting any particular metric or threshold. It depends only on the court accepting that (1) the BISECT maps are genuinely neutral (demonstrable by the algorithm’s lack of partisan data inputs), and (2) the enacted maps are further from the neutral centroid than natural geographic sorting explains (demonstrated by the six-metric table). The convergence of all six metrics in the same direction, and the consistency of the finding across four states, makes metric-shopping criticism untenable: the result is not an artifact of metric selection.

9 Legal Landscape and State Court Pathways

10 Legal Landscape and State Court Pathways

10.1 Post-*Rucho* Federal Foreclosure

Rucho v. Common Cause (2019) forecloses federal judicial review of partisan gerrymandering claims under the Constitution. The holding rests on two grounds: (1) no judicially manageable standard exists to distinguish permissible partisan line-drawing from unconstitutional

excess, and (2) partisan gerrymandering claims present political questions that federal courts lack the institutional competence to resolve. The majority’s survey of proposed standards—equal protection, First Amendment association, Elections Clause—found each either too broad (invalidating all partisan redistricting) or too indeterminate (requiring courts to set the acceptable level of partisanship, a choice the majority characterized as “not law” but politics).⁹

This forecloses not just claims under the Constitution’s partisan gerrymandering doctrine but also the federal common law of redistricting as a check on partisan maps. Federal courts can still review congressional redistricting for *Wesberry* population equality violations, *Shaw v. Reno* racial gerrymandering, and Voting Rights Act compliance—but not partisan composition.

10.2 State Constitutional Pathways

Three active state constitutional pathways have produced successful partisan gerrymandering claims since *Rucho*:

Pennsylvania’s Free and Equal Elections Clause (Art. I, §5). The Pennsylvania Supreme Court, in *League of Women Voters v. Commonwealth* (2018)—decided before *Rucho*, but under a state constitutional theory immune to federal foreclosure—held that Pennsylvania’s congressional map violated the free and equal elections clause. The court drew a remedial map itself, applying compactness, contiguity, and population equality criteria without partisan data. The EG and MM of the court-drawn remedial map were substantially lower than the invalidated enacted map, demonstrating that state constitutional standards can be operationalized through metric evidence.

North Carolina’s Free Elections Clause (Art. I, §10). In *Harper v. Hall* (N.C. 2022), the North Carolina Supreme Court held that the legislature “may not draw districts for the purpose of systematically advantaging or disadvantaging a particular political party” and that the 2022 congressional map violated this standard. The court relied on multiple partisan fairness metrics—including EG and MM—as evidence that the legislature had subordinated all legitimate redistricting criteria to partisan advantage. The court did not adopt a specific numerical threshold for any metric; instead it treated the convergence of multiple metrics pointing in the same direction as evidence of constitutional violation.

Ohio’s Anti-Gerrymandering Provision (Ohio Const. Art. XI, §6). The Ohio Constitution’s 2015 amendment explicitly prohibits drawing district boundaries “for the purpose of favoring or disfavoring a political party.” In *League of Women Voters of Ohio v. Ohio*

⁹*Rucho*, 588 U.S. at 720–21.

Redistricting Commission (2022), the Ohio Supreme Court applied this standard to invalidate the Ohio congressional map seven times over three years as the legislature repeatedly submitted plans that failed the constitutional test. The court’s analysis included seat-share projections and vote-share analysis that map onto the partisan metrics in this paper.

10.3 The Metric Selection Problem in State Court Practice

Expert witnesses in state redistricting litigation face a metric selection problem: which of the six metrics provides the strongest, most defensible evidence under the applicable state constitutional standard?

The answer depends on the legal theory. For NC’s “free elections” clause, which has been interpreted to prohibit systematic partisan advantage rather than requiring proportionality, symmetry-based metrics (Partisan Bias, EG) and the seats-votes curve are most directly relevant—they measure whether the map treats the two parties symmetrically when their vote totals are equal. For PA’s free and equal elections clause, which has been interpreted to impose a softer proportionality floor, the Gallagher Index and MM are also relevant. For Ohio’s explicit anti-gerrymandering provision, the full six-metric package demonstrates systematic partisan intent most completely.

The decision table in Section 4 (Table 1) operationalizes this guidance. Expert witnesses should lead with the metric most directly tied to the applicable legal standard, report all six for convergence, and explain that the BISECT reference distribution establishes the neutral baseline from which the enacted map’s deviation is measured. Metric-shopping criticism—the objection that the expert selected the metric most favorable to their client—is defeated by reporting all six simultaneously and showing that all six converge.

10.4 What Courts Have and Have Not Adopted

Through 2026, the following metric adoptions and rejections can be documented:

- **EG:** Cited in *Gill v. Whitford* (federal, not adopted as standard), *Whitford v. Gill* (7th Cir., applied but not dispositive), *Harper v. Hall* (NC, treated as one of several indicators).
- **MM:** Cited in *Harper v. Hall* (NC), academic literature widely. No court has adopted a specific MM threshold.
- **Partisan Bias:** Part of the symmetry framework cited in NC and PA expert testimony. Courts have treated it as relevant evidence but not dispositive.

- **Declination:** Introduced as an exhibit in *Common Cause v. Rucho* (M.D.N.C. 2018, pre-*Rucho* SCOTUS); the court did not rely on it. No federal or state court has adopted declination as evidence as of 2026 (see L.4).
- **Seats-votes curve:** Admitted as expert evidence in NC and OH proceedings. Courts have used the graphical exhibit without adopting a specific responsiveness threshold.
- **Gallagher LSq:** Used in academic analysis of redistricting; not adopted as a standalone legal standard in any U.S. court as of 2026.

11 Conclusion

12 Conclusion

This paper has provided a unified framework for the six partisan fairness metrics that practitioners must navigate in post-*Rucho* redistricting litigation. The central contributions are three.

First, the taxonomy of descriptive vs. normative metrics clarifies what each measure captures and what it cannot. Descriptive metrics (EG, MM, Bias, Declination) characterize the partisan properties of a given map; they do not specify what the properties should be. Normative metrics (proportionality, responsiveness) impose a target; they require a prior normative commitment that courts must make explicitly. Confusing these categories—treating a descriptive metric as if it contained its own standard, or treating a normative metric as a value-free measurement—produces avoidable expert testimony errors.

Second, the decision table (Table 1) maps state constitutional legal theory to metric selection, reducing the risk of metric-shopping criticism. Expert witnesses should lead with the metric most directly relevant to the applicable state constitutional standard, report all six for convergence, and explain the relationships among them to preempt opposing arguments that any single number was cherry-picked.

Third, the BISECT reference distribution provides state courts with a conceptually clear, evidentiary tool for measuring the magnitude of partisan manipulation. Because BISECT maps cannot gerrymander—they lack the partisan data inputs that gerrymandering requires—their partisan properties reflect geographic sorting alone. The gap between BISECT values and enacted map values on any metric measures the portion of partisan skew that is attributable to deliberate mapmaker choices rather than residential patterns. For North Carolina, this gap is $3.5\times$ to $7\times$ larger than what geography alone produces, varying by metric; efficiency

gap and mean-median ratios are near $3.5\times$, while partisan bias ratios reach $7\times$ in enacted maps with extreme manipulation.

The L.1–L.6 companion papers provide dedicated treatments of each metric, including the full mathematical development, empirical results across NC, WI, TX, and FL, legal history, and known limitations. Courts, commissioners, and expert witnesses should use this overview paper to orient the framework and consult the dedicated papers for the depth of analysis required in contested proceedings.

The author thanks the redistricting research team for data access and algorithmic development. All errors are the author’s own.

References